Task 2

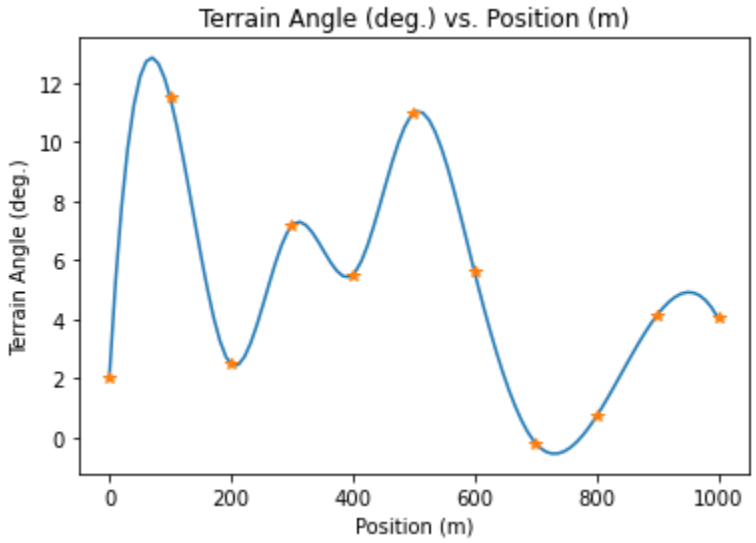


Figure 1. Terrain Angle vs. Position

In figure 1, it is seen that as the rover drives from 0 to 1000 m, it goes over steeper regions specifically from 0 to 200, and 400 to 600 being the steepest.The rover goes downhill whenever the angles are below zero so from around 700 to 750 m. There are no sharp corners in this graph because the graph has a continuous slope. All the data points being marked with a star, are on the curve due to interpolation. For interpolation, it is required for the curve to pass through every data point because the data points provided are the only exact values in this method. Nevertheless, the maximum and minimum values of the actual data and the curve differ. For the curve, the highest value happens at around 70 m with an angle of around 14 degrees and the lowest would be around -1 degree at around 700 m. In contrast, the highest data value is 11.5 degrees at 100 m and the smallest is -0.18 degrees at 700 m. This happens because the curve uses the other values that are not the provided data as estimates, this in order to stay smooth.

Task 5

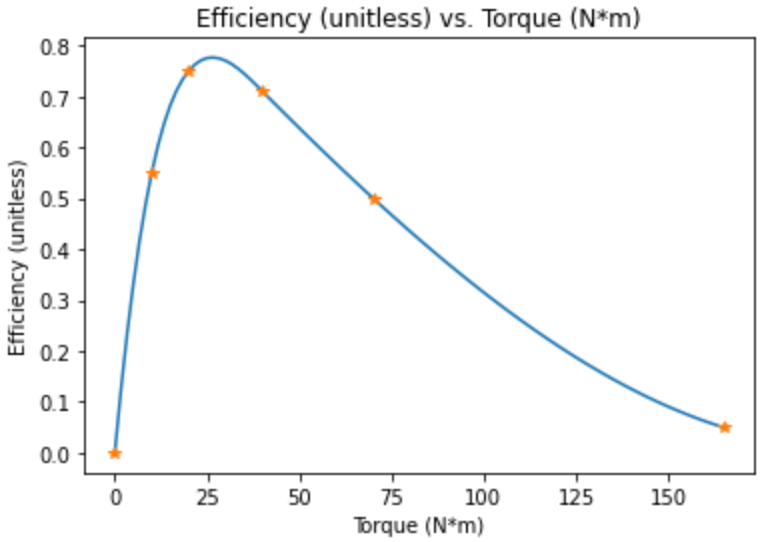


Figure 2. Efficiency vs. Torque

In Figure 2, it is seen that the maximum data point does not match with the peak of the curve. In the curve, the efficiency drastically increases until it reaches its peak at round 25 N*m with an efficiency of around .78 while the maximum efficiency provided is .75 at 20 N*m. The curve passes through every data point market with a star because interpolation is being used, which requires calculating estimations for values between the provided ones. While the provided data points are treated as exact and are being connected through a smooth continuous curve with a slope that is continuous at each data point.

In Figure 2 of the handout, it is seen that the efficiency curve of a typical DC motor is similar to Figure 2 of this document. Where the shape is similar, with an early peak. It is important to note that the units of Figure 2 of the handout are in percentages, where it gives a peak at around 76% at around .2 N*m of torque. An important difference on both graphs is the decline portion, where the decline of the handout's graph is more straight while the one in Figure 2 is curved. This could be because the graph from Figure 2 had a limited amount of data points.

Task 8

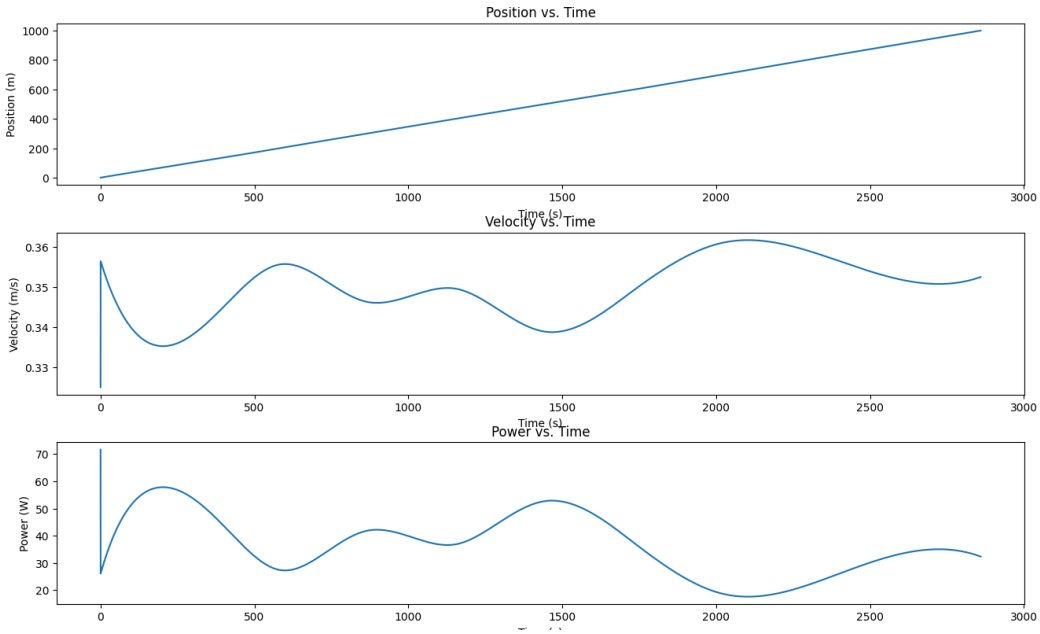


Figure 3. Position vs. Time (Top), Velocity vs. Time (Middle), Power vs. Time (Bottom)

In figure 3, it is seen that the Position vs Time graph is almost linear, meaning the rover's position increases at a nearly steady rate that gets affected by the terrain angle. The velocity and power graphs have similar shapes, but inverse of each other. This makes sense because the power depends on the motor torque and angular velocity. For example, if the rover is going uphill, it will be going at a slower rate, requiring more torque and thus more power. On the other hand, if the rover is going downhill, it would be the opposite, where it will be going at a faster rate, requiring less torque and power.

In Figure 3, the velocity profile is related to the terrain angle in Figure 1 because the velocity decreases at 200 seconds, where the rover is at 100 m and the terrain angle is increasing. This makes sense because when the terrain angle increases, the rover will be travelling uphill and there will be more gravitational force opposing its movement, decreasing the rover's velocity. The velocity graph is smooth because the rover's velocity changes gradually in response to the forces acting on the rover as it travels across the terrain.

Table 1. Values of Completion Time, Distance Traveled, Max Velocity, Average Velocity, Battery Energy, and Battery Energy per Distance.

Field	Values
completion_time	2859.48 s
distance_traveled	1000 m
max_velocity	0.3618 m/s

average_velocity	0.3497 m/s
battery_energy	1202408.32 J
batt_energy_per_distance	1202.408 J/m

Task 9

The battery energy required for experiment 1 was 1022408.32 J. The Lithium Iron Phosphate battery pack would not allow for the rover to complete the case from Task 8 because it does not have enough with only 0.9072e6 J of energy.